

## Ultimate Nutrition

### Liquid Amino

Ultimate Nutrition introduces New Liquid Amino; a fast-acting, easily-digested liquid concentrate that provides a full spectrum of essential and non-essential amino acids. Each serving contains 22 grams of high-quality protein, derived from natural enzyme-hydrolyzed gelatin and whey protein isolate. Amino acids are the building blocks of protein in the body; as such they are essential for the synthesis of structural proteins, enzymes, and some hormones and neurotransmitters. Amino acids are also involved in numerous metabolic pathways that affect exercise metabolism.

Athletes are accustomed to viewing carbohydrates as the “exercise macronutrient”. Carbohydrates, primarily in the form of muscle glycogen, supply the majority of fuel at the beginning of high-intensity training. However, amino acids, which are supplied via breakdown of muscle protein, contribute an increasing amount of energy during prolonged endurance activity. Infact, research has shown that exercise burns up to 15% of the total amount of calories from protein by extracting amino acids from muscle tissues. If athletes do not supply this protein as part of their diet, lean muscle tissue will be used. Without amino acid replenishment, as athletes exercise longer, their performance, recovery, and immune system are compromised.

Liquid Amino is a combination of proteins that provide complete support for an athlete’s diet. Each delicious Fruit Punch flavored serving provides 22 grams of high quality protein derived from natural enzyme hydrolyzed gelatin and whey protein isolate.

Whey protein is excellent for enhancing recovery. Whey protein has the highest biological value (BV) of any protein source. Whey is also a rich source of two important amino acids, methionine and cysteine, which stimulate the natural production of glutathione, one of the body’s most powerful antioxidants and a major factor in maintaining a strong immune system. Glutathione also supports healthy liver function.

Bovine gelatin has been marketed worldwide for many years as a food and as a protein and joint health supplement. Collagen is an essential structural protein that forms an important part of bones, tendons, and connective tissues. It is a tough insoluble protein that is essential for keeping the many cells and tissues of the body together. Gelatin contains an exceptionally high content of two amino acids, which play an important part in collagen formation; proline and glycine. The intake of hydrolyzed gelatin gets chondrocytes and osteoblasts (cartilage producing cells and bone forming cells) ample amounts of these important amino acids for making structural proteins.

Ultimate Nutrition’s Liquid Amino is a fast-acting, easily-digested liquid concentrate providing a full spectrum of essential and non-essential amino acids. Each delicious Fruit Punch flavored serving provides 22 grams of high-quality protein derived from natural enzyme hydrolyzed gelatin and whey protein isolate.

## SELECTED REFERENCES:

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## FREQUENTLY ASKED QUESTIONS:

1. Does Liquid Amino need to be mixed?

No. Liquid Amino is in a delicious Fruit Punch flavored liquid form that is ready to use. Each 50 mL serving contains 22 grams of protein.

2. Can I store Liquid Amino in the refrigerator?

It is not recommended. Liquid Amino is best stored in a cool dry place. If placed in the refrigerator it will thicken, making it difficult to pour and measure.

3. Do all amino acids need to be taken in the same amounts?

No. Different amino acids should be taken in different amounts. The ratios depend upon the relative natural abundance of each amino acid in the body and amount required for incorporation in proteins.

4. Why are BCAAs beneficial?

BCAAs (Branched Chain Amino Acids) are the first amino acids to be catabolized due to intense exercise. The body requires higher amounts of branched chain amino acids during and following exercise as they are taken up directly by the skeletal muscles versus first being metabolized through the liver, like other amino acids. Low BCAA levels contribute to fatigue and they should be replaced in one-hour or less following exercise.